



《Linear Model》
Chapter 4
Multiple Regression: Estimation
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Chapter 4 Multiple Regression: Estimation

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4.1 THE MODEL

The multiple linear regression model, as introduced in Section 1.2, can be expressed as

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon. \quad (7.1)$$

We discuss estimation of the β parameters when the model is linear in the β 's. An example of a model that is linear in the β 's but not the x 's is the second-order response surface model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2 + \beta_5 x_1 x_2 + \varepsilon. \quad (7.2)$$

4.1 THE MODEL

To estimate the β 's in (7.1), we will use a sample of n observations on y and the associated x variables. The model for the i th observation is

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i, \quad i = 1, 2, \dots, n. \quad (7.3)$$

The assumptions for ε_i or y_i are essentially the same as those for simple linear regression in Section 6.1:

1. $E(\varepsilon_i) = 0$ for $i = 1, 2, \dots, n$, or, equivalently, $E(y_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik}$.
2. $\text{var}(\varepsilon_i) = \sigma^2$ for $i = 1, 2, \dots, n$, or, equivalently, $\text{var}(y_i) = \sigma^2$.
3. $\text{cov}(\varepsilon_i, \varepsilon_j) = 0$ for all $i \neq j$, or, equivalently, $\text{cov}(y_i, y_j) = 0$.

4.1 THE MODEL

Writing (7.3) for each of the n observations, we have

$$y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \cdots + \beta_k x_{1k} + \varepsilon_1$$

$$y_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \cdots + \beta_k x_{2k} + \varepsilon_2$$

$$\vdots$$

$$y_n = \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \cdots + \beta_k x_{nk} + \varepsilon_n.$$

4.1 THE MODEL

These n equations can be written in matrix form as

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

or

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}. \quad (7.4)$$

4.1 THE MODEL

The preceding three assumptions on ε_i or y_i can be expressed in terms of the model in (7.4):

1. $E(\varepsilon) = \mathbf{0}$ or $E(y) = \mathbf{X}\beta$.
2. $\text{cov}(\varepsilon) = \sigma^2 \mathbf{I}$ or $\text{cov}(y) = \sigma^2 \mathbf{I}$.

Note that the assumption $\text{cov}(\varepsilon) = \sigma^2 \mathbf{I}$ includes both the previous assumptions $\text{var}(\varepsilon_i) = \sigma^2$ and $\text{cov}(\varepsilon_i, \varepsilon_j) = 0$.

The matrix $\mathbf{X}$ in (7.4) is $n \times (k + 1)$. In this chapter we assume that $n > k + 1$ and $\text{rank}(\mathbf{X}) = k + 1$. If $n < k + 1$ or if there is a linear relationship among the x 's, for example, $x_5 = \sum_{j=1}^4 x_j/4$, then $\mathbf{X}$ will not have full column rank. If the values of the x_{ij} 's are planned (chosen by the researcher), then the $\mathbf{X}$ matrix essentially contains the experimental design and is sometimes called the *design matrix*.

4.1 THE MODEL

The β parameters in (7.1) or (7.4) are called *regression coefficients*. To emphasize their collective effect, they are sometimes referred to as *partial regression coefficients*. The word *partial* carries both a mathematical and a statistical meaning. Mathematically, the partial derivative of $E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k$ with respect to x_1 , for example, is β_1 . Thus β_1 indicates the change in $E(y)$ with a unit increase in x_1 when $x_2, x_3, \dots, x_k$ are held constant. Statistically, β_1 shows the effect of x_1 on $E(y)$ in the presence of the other x 's. This effect would typically be different from the effect of x_1 on $E(y)$ if the other x 's were not present in the model. Thus, for example, β_0 and β_1 in

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

will usually be different from β_0^* and β_1^* in

$$y = \beta_0^* + \beta_1^* x_1 + \varepsilon^*.$$

4.2 ESTIMATION OF β AND σ^2

Least-Squares Estimator for β

For the parameters $\beta_0, \beta_1, \dots, \beta_k$, we seek estimators that minimize the sum of squares of deviations of the n observed y 's from their predicted values $\hat{y}$. By extension of (6.2), we seek $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ that minimize

$$\begin{aligned}\sum_{i=1}^n \hat{\varepsilon}_i^2 &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2} - \dots - \hat{\beta}_k x_{ik})^2.\end{aligned}\quad (7.5)$$

Note that the predicted value $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_k x_{ik}$ estimates $E(y_i)$, not y_i . A better notation would be $\widehat{E(y_i)}$, but $\hat{y}_i$ is commonly used.

To obtain the least-squares estimators, it is not necessary that the prediction equation $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_k x_{ik}$ be based on $E(y_i)$. It is only necessary to postulate an empirical model that is linear in the $\hat{\beta}$'s, and the least-squares method will find the "best" fit to this model. This was illustrated in Figure 6.2.

4.2 ESTIMATION OF β AND σ^2

To find the values of $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ that minimize (7.5), we could differentiate $\sum_i \hat{\varepsilon}_i^2$ with respect to each $\hat{\beta}_j$ and set the results equal to zero to yield $k + 1$ equations that can be solved simultaneously for the $\hat{\beta}_j$'s. However, the procedure can be carried out in more compact form with matrix notation. The result is given in the following theorem.

Theorem 7.3a. If $\mathbf{y} = \mathbf{X}\beta + \varepsilon$, where $\mathbf{X}$ is $n \times (k + 1)$ of rank $k + 1 < n$, then the value of $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k)'$ that minimizes (7.5) is

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{y}.\quad (7.6)$$

4.2 ESTIMATION OF β AND σ^2 

一些矩阵微商公式表

$y = f(X)$	$\frac{\partial y}{\partial X}$
$a'x$	a
$x'Ax$	$2Ax$
$ X $	$\begin{cases} X (X^{-1})', \\ X (2X^{-1} - \text{diag}(X^{-1})) \end{cases} \quad (X \text{ 对称})$
$ AXB $	$ AXB A'((AXB)^{-1})'B'$
$\ln AXB $	$A'((AXB)^{-1})'B'$
$\ln X'AX $	$2AX(X'AX)^{-1} \quad (A \text{ 对称})$
$\text{tr}(XAX')$	$X(A + A')$
$\text{tr}(AXB)$	$A'B'$
$\text{tr}(X'AXB)$	$AXB + A'XB'$
$\ln X $	$2X^{-1} - \text{diag}(X^{-1}) \quad (X \text{ 对称})$
$\text{tr}(AX)$	$A + A' - \text{diag}(A) \quad (X \text{ 对称})$

4.2 ESTIMATION OF β AND σ^2 

PROOF. Using (2.20) and (2.27), we can write (7.5) as

$$\hat{\varepsilon}'\hat{\varepsilon} = \sum_{i=1}^n (y_i - \mathbf{x}_i'\hat{\beta})^2 = (\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta}), \quad (7.7)$$

where $\mathbf{x}_i' = (1, x_{i1}, \dots, x_{ik})$ is the i th row of $\mathbf{X}$. When the product $(\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta})$ in (7.7) is expanded as in (2.17), two of the resulting four terms can be combined to yield

$$\hat{\varepsilon}'\hat{\varepsilon} = \mathbf{y}'\mathbf{y} - 2\mathbf{y}'\mathbf{X}\hat{\beta} + \hat{\beta}'\mathbf{X}'\mathbf{X}\hat{\beta}.$$

We can find the value of $\hat{\beta}$ that minimizes $\hat{\varepsilon}'\hat{\varepsilon}$ by differentiating $\hat{\varepsilon}'\hat{\varepsilon}$ with respect to $\hat{\beta}$ [using (2.112) and (2.113)] and setting the result equal to zero:

$$\frac{\partial \hat{\varepsilon}'\hat{\varepsilon}}{\partial \hat{\beta}} = \mathbf{0} - 2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{0},$$

4.2 ESTIMATION OF β AND σ^2

This gives the *normal equations*

$$\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}. \quad (7.8)$$

By Theorems 2.4(iii) and 2.6d(i) and Corollary 1 of Theorem 2.6c, if $\mathbf{X}$ is full-rank, $\mathbf{X}'\mathbf{X}$ is nonsingular, and the solution to (7.8) is given by (7.6). $\square$

Since $\hat{\beta}$ in (7.6) minimizes the sum of squares in (7.5), $\hat{\beta}$ is called the *least-squares estimator*. Note that each $\hat{\beta}_j$ in $\hat{\beta}$ is a linear function of $\mathbf{y}$; that is, $\hat{\beta}_j = \mathbf{a}'_j \mathbf{y}$, where $\mathbf{a}'_j$ is the j th row of $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. This usage of the word *linear* in *linear estimator* is different from that in *linear model*, which indicates that the model is linear in the β 's.

4.2 ESTIMATION OF β AND σ^2

We now show that $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ minimizes $\hat{\epsilon}'\hat{\epsilon}$. Let $\mathbf{b}$ be an alternative estimator that may do better than $\hat{\beta}$ so that $\hat{\epsilon}'\hat{\epsilon}$ is

$$\hat{\epsilon}'\hat{\epsilon} = (\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b}).$$

Now adding and subtracting $\mathbf{X}\hat{\beta}$, we obtain

$$= (\mathbf{y} - \mathbf{X}\hat{\beta} + \mathbf{X}\hat{\beta} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\hat{\beta} + \mathbf{X}\hat{\beta} - \mathbf{X}\mathbf{b}) \quad (7.9)$$

$$= (\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta}) + (\hat{\beta} - \mathbf{b})'\mathbf{X}'\mathbf{X}(\hat{\beta} - \mathbf{b}) + 2(\hat{\beta} - \mathbf{b})'(\mathbf{X}'\mathbf{y} - \mathbf{X}'\mathbf{X}\hat{\beta}). \quad (7.10)$$

The third term on the right side of (7.10) vanishes because of the normal equations $\mathbf{X}'\mathbf{y} = \mathbf{X}'\mathbf{X}\hat{\beta}$ in (7.8). The second term is a positive definite quadratic form (assuming that $\mathbf{X}$ is full-rank; see Theorem 2.6d), and $\hat{\epsilon}'\hat{\epsilon}$ is therefore minimized when $\mathbf{b} = \hat{\beta}$.

4.2 ESTIMATION OF β AND σ^2

To examine the structure of $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}'\mathbf{y}$, note that by Theorem 2.2c(i), the $(k+1) \times (k+1)$ matrix $\mathbf{X}'\mathbf{X}$ can be obtained as products of columns of $\mathbf{X}$; similarly, $\mathbf{X}'\mathbf{y}$ contains products of columns of $\mathbf{X}$ and $\mathbf{y}$:

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} n & \sum_i x_{i1} & \sum_i x_{i2} & \cdots & \sum_i x_{ik} \\ \sum_i x_{i1} & \sum_i x_{i1}^2 & \sum_i x_{i1}x_{i2} & \cdots & \sum_i x_{i1}x_{ik} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum_i x_{ik} & \sum_i x_{i1}x_{ik} & \sum_i x_{i2}x_{ik} & \cdots & \sum_i x_{ik}^2 \end{pmatrix} \quad \mathbf{X}'\mathbf{y} = \begin{pmatrix} \sum_i y_i \\ \sum_i x_{i1}y_i \\ \vdots \\ \sum_i x_{ik}y_i \end{pmatrix}.$$

If $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ as in (7.6), then

$$\hat{\epsilon} = \mathbf{y} - \mathbf{X}\hat{\beta} = \mathbf{y} - \hat{\mathbf{y}} \quad (7.11)$$

is the vector of *residuals*, $\hat{\epsilon}_1 = y_1 - \hat{y}_1$, $\hat{\epsilon}_2 = y_2 - \hat{y}_2$, $\dots$, $\hat{\epsilon}_n = y_n - \hat{y}_n$. The residual vector $\hat{\epsilon}$ estimates ϵ in the model $\mathbf{y} = \mathbf{X}\beta + \epsilon$ and can be used to check the validity of the model and attendant assumptions; see Chapter 9.

4.2 ESTIMATION OF β AND σ^2

$$\mathbf{y} = \begin{pmatrix} 2 \\ 3 \\ 2 \\ 7 \\ 6 \\ 8 \\ 10 \\ 7 \\ 8 \\ 12 \\ 11 \\ 14 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & 0 & 2 \\ 1 & 2 & 6 \\ 1 & 2 & 7 \\ 1 & 2 & 5 \\ 1 & 4 & 9 \\ 1 & 4 & 8 \\ 1 & 4 & 7 \\ 1 & 6 & 10 \\ 1 & 6 & 11 \\ 1 & 6 & 9 \\ 1 & 8 & 15 \\ 1 & 8 & 13 \end{pmatrix}.$$

```

y<-c(2, 3, 2, 7, 6, 8, 10, 7, 8, 12, 11,14)
x<-
matrix(c(rep(1,12),0,2,2,2,4,4,4,6,6,6,8,8,2,6,7,5,9,8,7,1
0,11,9,15,3),12,3)
> xtx
      [,1] [,2] [,3]
[1,]   12    52    92
[2,]   52   296   456
[3,]   92   456   844
> xty
      [,1]
[1,]    90
[2,]   482
[3,]   732
> xtxinv<-solve(xtx)
> xtxinv
      [,1]      [,2]      [,3]
[1,]  0.53603604 -0.024774775 -0.045045045
[2,] -0.02477477  0.021294021 -0.008804259
[3,] -0.04504505 -0.008804259  0.010851761
> beta<-xtxinv%*%xty
> beta
      [,1]
[1,]  3.3288288
[2,]  1.5892711
[3,] -0.3542179

```


4.2 ESTIMATION OF β AND σ^2 Properties of the Least-Squares Estimator $\hat{\beta}$ **Theorem 7.3b.** If $E(y) = X\beta$, then $\hat{\beta}$ is an unbiased estimator for β .

PROOF

$$\begin{aligned}
 E(\hat{\beta}) &= E[(X'X)^{-1}X'y] \\
 &= (X'X)^{-1}X'E(y) \quad [\text{by (3.38)}] \\
 &= (X'X)^{-1}X'X\beta \\
 &= \beta.
 \end{aligned} \tag{7.13}$$

4.2 ESTIMATION OF β AND σ^2 **Theorem 7.3c.** If $\text{cov}(y) = \sigma^2 I$, the covariance matrix for $\hat{\beta}$ is given by $\sigma^2(X'X)^{-1}$.

PROOF

$$\begin{aligned}
 \text{cov}(\hat{\beta}) &= \text{cov}[(X'X)^{-1}X'y] \\
 &= (X'X)^{-1}X'\text{cov}(y)[(X'X)^{-1}X']' \quad [\text{by (3.44)}] \\
 &= (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} \\
 &= \sigma^2(X'X)^{-1}X'X(X'X)^{-1} \\
 &= \sigma^2(X'X)^{-1}.
 \end{aligned} \tag{7.14}$$

4.2 ESTIMATION OF β AND σ^2



Theorem 7.3d (Gauss–Markov Theorem). If $E(y) = X\beta$ and $\text{cov}(y) = \sigma^2 I$, the least-squares estimators $\hat{\beta}_j, j = 0, 1, \dots, k$, have minimum variance among all linear unbiased estimators.

The Gauss–Markov theorem is sometimes stated as follows. If $E(y) = X\beta$ and $\text{cov}(y) = \sigma^2 I$, the least-squares estimators $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ are *best linear unbiased estimators* (BLUE). In this expression, *best* means minimum variance and *linear* indicates that the estimators are linear functions of y .

The remarkable feature of the Gauss–Markov theorem is its distributional generality. The result holds for any distribution of y ; normality is not required. The only assumptions used in the proof are $E(y) = X\beta$ and $\text{cov}(y) = \sigma^2 I$. If these assumptions do not hold, $\hat{\beta}$ may be biased or each $\hat{\beta}_j$ may have a larger variance than that of some other estimator.

4.2 ESTIMATION OF β AND σ^2



The Gauss–Markov theorem is easily extended to a linear combination of the $\hat{\beta}$'s, as follows.

Corollary 1. If $E(y) = X\beta$ and $\text{cov}(y) = \sigma^2 I$, the best linear unbiased estimator of $a'\beta$ is $a'\hat{\beta}$, where $\hat{\beta}$ is the least-squares estimator $\hat{\beta} = (X'X)^{-1}X'y$.

4.2 ESTIMATION OF β AND σ^2

An Estimator for σ^2

The method of least squares does not yield a function of the y and x values in the sample that we can minimize to obtain an estimator of σ^2 . However, we can devise an unbiased estimator for σ^2 based on the least-squares estimator $\hat{\beta}$. By assumption 2 following (7.3), σ^2 is the same for each y_i , $i = 1, 2, \dots, n$. By (3.6), σ^2 is defined by $\sigma^2 = E[y_i - E(y_i)]^2$, and by assumption 1, we obtain

$$E(y_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} = \mathbf{x}_i' \beta,$$

where $\mathbf{x}_i'$ is the i th row of $\mathbf{X}$. Thus σ^2 becomes

$$\sigma^2 = E[v_i - \mathbf{x}_i' \beta]^2.$$

We estimate σ^2 by a corresponding average from the sample

$$s^2 = \frac{1}{n - k - 1} \sum_{i=1}^n (y_i - \mathbf{x}_i' \hat{\beta})^2, \quad (7.22)$$

where n is the sample size and k is the number of x 's. Note that, by the corollary to Theorem 7.3d, $\mathbf{x}_i' \hat{\beta}$ is the BLUE of $\mathbf{x}_i' \beta$.

4.2 ESTIMATION OF β AND σ^2

Using (7.7), we can write (7.22) as

$$s^2 = \frac{1}{n - k - 1} (\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta}) \quad (7.23)$$

$$= \frac{\mathbf{y}'\mathbf{y} - \hat{\beta}'\mathbf{X}'\mathbf{y}}{n - k - 1} = \frac{\text{SSE}}{n - k - 1}, \quad (7.24)$$

Note: $E(\mathbf{y} - \mathbf{X}\hat{\beta}) = 0$

where $\text{SSE} = (\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathbf{y}'\mathbf{y} - \hat{\beta}'\mathbf{X}'\mathbf{y}$. With the denominator $n - k - 1$, s^2 is an unbiased estimator of σ^2 , as shown below.

Theorem 7.3f. If s^2 is defined by (7.22), (7.23), or (7.24) and if $E(\mathbf{y}) = \mathbf{X}\beta$ and $\text{cov}(\mathbf{y}) = \sigma^2 \mathbf{I}$, then

$$E(s^2) = \sigma^2. \quad (7.25)$$

4.2 ESTIMATION OF β AND σ^2

PROOF. Using (7.24) and (7.6), we write SSE as a quadratic form:

$$\begin{aligned} \text{SSE} &= \mathbf{y}'\mathbf{y} - \hat{\beta}'\mathbf{X}'\mathbf{y} = \mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\ &= \mathbf{y}'[\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{y}. \end{aligned} \quad (7.26)$$

By Theorem 5.2a, we have

$$\begin{aligned} E(\text{SSE}) &= \text{tr}\{[\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\sigma^2\mathbf{I}\} &&= \sigma^2\text{tr}[\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'] \\ &+ E(\mathbf{y}')[\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']E(\mathbf{y}) &&+ \beta'\mathbf{X}'[\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{X}\beta \\ &= \sigma^2\{n - \text{tr}[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\} &&= \sigma^2\{n - \text{tr}[\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}]\} \\ &+ \beta'\mathbf{X}'\mathbf{X}\beta - \beta'\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta &&+ \beta'\mathbf{X}'\mathbf{X}\beta - \beta'\mathbf{X}'\mathbf{X}\beta \quad [\text{by (2.87)}]. \end{aligned}$$

Since $\mathbf{X}'\mathbf{X}$ is $(k+1) \times (k+1)$, this becomes

$$E(\text{SSE}) = \sigma^2[n - \text{tr}(\mathbf{I}_{k+1})] = \sigma^2(n - k - 1).$$

4.2 ESTIMATION OF β AND σ^2

Corollary 1. An unbiased estimator of $\text{cov}(\hat{\beta})$ in (7.14) is given by

$$\widehat{\text{cov}}(\hat{\beta}) = s^2(\mathbf{X}'\mathbf{X})^{-1}. \quad (7.27)$$

Example 7.3.3. For the data in Table 7.1, we have

$$\text{SSE} = \mathbf{y}'\mathbf{y} - \hat{\beta}'\mathbf{X}\mathbf{y}$$

$$s^2 = \frac{\text{SSE}}{n - k - 1}$$

```
SSE=t(y)%*%y-t(beta)%*%xty
s2=SSE/(12-2-1)
```

```
> SSE=t(y)*%y-t(beta)*%xty
> SSE
```

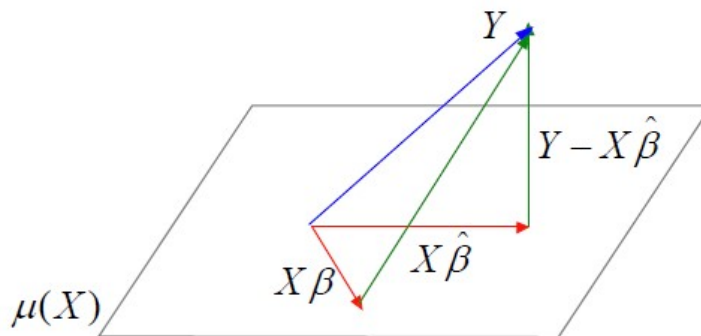
```
      [,1]
[1,] 33.66421
> s2=SSE/(12-2-1)
> s2
```

```
      [,1]
[1,] 3.740468
```

4.3 Geometric Interpretation of the Multiple Linear Regression Model



$$\min_{\beta} \|Y - X\beta\|^2 = \min_{\theta \in \mu(X)} \|Y - \theta\|^2$$



$$HY = X(X'X)^{-1}X'Y = X\hat{\beta}$$

4.4 THE MODEL IN CENTERED FORM



The model in (7.3) for each y_i can be written in terms of centered x variables as

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i \\ &= \alpha + \beta_1 (x_{i1} - \bar{x}_1) + \beta_2 (x_{i2} - \bar{x}_2) + \cdots + \beta_k (x_{ik} - \bar{x}_k) + \varepsilon_i, \end{aligned} \quad (7.30)$$

$i = 1, 2, \dots, n$, where

$$\alpha = \beta_0 + \beta_1 \bar{x}_1 + \beta_2 \bar{x}_2 + \cdots + \beta_k \bar{x}_k \quad (7.31)$$

and $\bar{x}_j = \sum_{i=1}^n x_{ij}/n, j = 1, 2, \dots, k$. The centered form of the model is useful in expressing certain hypothesis tests (Section 8.1), in a search for influential observations (Section 9.2), and in providing other insights.

4.4 THE MODEL IN CENTERED FORM

In matrix form, the centered model (7.30) for $y_1, y_2, \dots, y_n$ becomes

$$\mathbf{y} = (\mathbf{j}, \mathbf{X}_c) \begin{pmatrix} \alpha \\ \boldsymbol{\beta}_1 \end{pmatrix} + \boldsymbol{\varepsilon}, \quad (7.32)$$

where $\boldsymbol{\beta}_1 = (\beta_1, \beta_2, \dots, \beta_k)'$,

$$\mathbf{X}_c = \left(\mathbf{I} - \frac{1}{n} \mathbf{J} \right) \mathbf{X}_1 = \begin{pmatrix} x_{11} - \bar{x}_1 & x_{12} - \bar{x}_2 & \dots & x_{1k} - \bar{x}_k \\ x_{21} - \bar{x}_1 & x_{22} - \bar{x}_2 & \dots & x_{2k} - \bar{x}_k \\ \vdots & \vdots & & \vdots \\ x_{n1} - \bar{x}_1 & x_{n2} - \bar{x}_2 & \dots & x_{nk} - \bar{x}_k \end{pmatrix}, \quad (7.33)$$

and $\mathbf{X}_1$ is as given in (7.19). The matrix $\mathbf{I} - (1/n)\mathbf{J}$ is sometimes called the *centering matrix*.

4.4 THE MODEL IN CENTERED FORM

As in (7.8), the normal equations for the model in (7.32) are

$$(\mathbf{j}, \mathbf{X}_c)'(\mathbf{j}, \mathbf{X}_c) \begin{pmatrix} \hat{\alpha} \\ \hat{\boldsymbol{\beta}}_1 \end{pmatrix} = (\mathbf{j}, \mathbf{X}_c)' \mathbf{y}. \quad (7.34)$$

By (2.35) and (2.39), the product $(\mathbf{j}, \mathbf{X}_c)'(\mathbf{j}, \mathbf{X}_c)$ on the left side of (7.34) becomes

$$\begin{aligned} (\mathbf{j}, \mathbf{X}_c)'(\mathbf{j}, \mathbf{X}_c) &= \begin{pmatrix} \mathbf{j}' \\ \mathbf{X}_c' \end{pmatrix} (\mathbf{j}, \mathbf{X}_c) = \begin{pmatrix} \mathbf{j}'\mathbf{j} & \mathbf{j}'\mathbf{X}_c \\ \mathbf{X}_c'\mathbf{j} & \mathbf{X}_c'\mathbf{X}_c \end{pmatrix} \\ &= \begin{pmatrix} n & \mathbf{0}' \\ \mathbf{0} & \mathbf{X}_c'\mathbf{X}_c \end{pmatrix}, \end{aligned} \quad (7.35)$$

where $\mathbf{j}'\mathbf{X}_c = \mathbf{0}'$ because the columns of $\mathbf{X}_c$ sum to zero (Problem 7.16). The right side of (7.34) can be written as

$$(\mathbf{j}, \mathbf{X}_c)' \mathbf{y} = \begin{pmatrix} \mathbf{j}' \\ \mathbf{X}_c' \end{pmatrix} \mathbf{y} = \begin{pmatrix} n\bar{y} \\ \mathbf{X}_c'\mathbf{y} \end{pmatrix}.$$

4.4 THE MODEL IN CENTERED FORM

The least-squares estimators are then given by

$$\begin{aligned} \begin{pmatrix} \hat{\alpha} \\ \hat{\beta}_1 \end{pmatrix} &= [(\mathbf{j}, \mathbf{X}_c)'(\mathbf{j}, \mathbf{X}_c)]^{-1}(\mathbf{j}, \mathbf{X}_c)'\mathbf{y} = \begin{pmatrix} n & \mathbf{0}' \\ \mathbf{0} & \mathbf{X}_c'\mathbf{X}_c \end{pmatrix}^{-1} \begin{pmatrix} n\bar{y} \\ \mathbf{X}_c'\mathbf{y} \end{pmatrix} \\ &= \begin{pmatrix} 1/n & \mathbf{0}' \\ \mathbf{0} & (\mathbf{X}_c'\mathbf{X}_c)^{-1} \end{pmatrix} \begin{pmatrix} n\bar{y} \\ \mathbf{X}_c'\mathbf{y} \end{pmatrix} = \begin{pmatrix} \bar{y} \\ (\mathbf{X}_c'\mathbf{X}_c)^{-1}\mathbf{X}_c'\mathbf{y} \end{pmatrix}, \end{aligned}$$

or

$$\hat{\alpha} = \bar{y}, \quad (7.36)$$

$$\hat{\beta}_1 = (\mathbf{X}_c'\mathbf{X}_c)^{-1}\mathbf{X}_c'\mathbf{y}. \quad (7.37)$$

4.4 THE MODEL IN CENTERED FORM

These estimators are the same as the usual least-squares estimators $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ in (7.6), with the adjustment

$$\hat{\beta}_0 = \hat{\alpha} - \hat{\beta}_1\bar{x}_1 - \hat{\beta}_2\bar{x}_2 - \cdots - \hat{\beta}_k\bar{x}_k = \bar{y} - \hat{\beta}'_1\bar{\mathbf{x}} \quad (7.38)$$

obtained from an estimator of α in (7.31) (see Problem 7.17).

When we express $\hat{\mathbf{y}}$ in centered form

$$\hat{\mathbf{y}} = \hat{\alpha} + \hat{\beta}_1(x_1 - \bar{x}_1) + \cdots + \hat{\beta}_k(x_k - \bar{x}_k),$$

it is clear that the fitted regression plane passes through the point $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_k, \bar{y})$.

Adapting the expression for SSE (7.24) to the centered model with centered $\hat{\mathbf{y}}$'s, we obtain

$$\text{SSE} = \sum_{i=1}^n (y_i - \bar{y})^2 - \hat{\beta}'_1 \mathbf{X}'_c \mathbf{y}, \quad (7.39)$$

which turns out to be equal to $\text{SSE} = \mathbf{y}'\mathbf{y} - \hat{\beta}'\mathbf{X}'\mathbf{y}$ (see Problem 7.19).

4.5 NORMAL MODEL

Assumptions

Thus far we have made no normality assumptions about the random variables $y_1, y_2, \dots, y_n$. To the assumptions in Section 7.2, we now add that

$$\mathbf{y} \text{ is } N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}) \quad \text{or} \quad \boldsymbol{\varepsilon} \text{ is } N_n(\mathbf{0}, \sigma^2\mathbf{I}).$$

Under normality, $\sigma_{ij} = 0$ implies that the y (or ε) variables are independent, as well as uncorrelated.

4.5 NORMAL MODEL

Maximum Likelihood Estimators for $\boldsymbol{\beta}$ and σ^2

Theorem 7.6a. If $\mathbf{y}$ is $N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$, where $\mathbf{X}$ is $n \times (k+1)$ of rank $k+1 < n$, the maximum likelihood estimators of $\boldsymbol{\beta}$ and σ^2 are

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}, \quad (7.48)$$

$$\hat{\sigma}^2 = \frac{1}{n}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}). \quad (7.49)$$

The maximum likelihood estimator $\hat{\boldsymbol{\beta}}$ in (7.48) is the same as the least-squares estimator $\hat{\boldsymbol{\beta}}$ in Theorem 7.3a. The estimator $\hat{\sigma}^2$ in (7.49) is biased since the denominator is n rather than $n - k - 1$. We often use the unbiased estimator s^2 given in (7.23) or (7.24).

4.5 NORMAL MODEL

PROOF. We sketch the proof. For the remaining steps, see Problem 7.21. The likelihood function (joint density of $y_1, y_2, \dots, y_n$) is given by the multivariate normal density (4.9)

$$\begin{aligned} L(\boldsymbol{\beta}, \sigma^2) = f(\mathbf{y}; \boldsymbol{\beta}, \sigma^2) &= \frac{1}{(2\pi)^{n/2} |\sigma^2 \mathbf{I}|^{1/2}} e^{-(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\sigma^2 \mathbf{I})^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})/2} \\ &= \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})/2\sigma^2}. \end{aligned} \quad (7.50)$$

[Since the y_i 's are independent, $L(\boldsymbol{\beta}, \sigma^2)$ can also be obtained as $\prod_{i=1}^n f(y_i; \mathbf{x}_i' \boldsymbol{\beta}, \sigma^2)$.] Then $\ln L(\boldsymbol{\beta}, \sigma^2)$ becomes

$$\ln L(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}). \quad (7.51)$$

Taking the partial derivatives of $\ln L(\boldsymbol{\beta}, \sigma^2)$ with respect to $\boldsymbol{\beta}$ and σ^2 and setting the results equal to zero will produce (7.48) and (7.49). To verify that $\hat{\boldsymbol{\beta}}$ maximizes (7.50) or (7.51), see (7.10). $\square$

4.5 NORMAL MODEL

Properties of $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$

Theorem 7.6b. Suppose that $\mathbf{y}$ is $N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$, where $\mathbf{X}$ is $n \times (k+1)$ of rank $k+1$, $1 < n$ and $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_k)'$. Then the maximum likelihood estimators $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ given in Theorem 7.6a have the following distributional properties:

- (i) $\hat{\boldsymbol{\beta}}$ is $N_{k+1}[\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}]$.
- (ii) $n\hat{\sigma}^2/\sigma^2$ is $\chi^2(n-k-1)$, or equivalently, $(n-k-1)s^2/\sigma^2$ is $\chi^2(n-k-1)$.
- (iii) $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ (or s^2) are independent.

PROOF

- (i) Since $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is a linear function of $\mathbf{y}$ of the form $\hat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{y}$, where $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is a constant matrix, then by Theorem 4.4a(ii), $\hat{\boldsymbol{\beta}}$ is $N_{k+1}[\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}]$.
- (ii) The result follows from Corollary 2 to Theorem 5.5.
- (iii) The result follows from Corollary 1 to Theorem 5.6a.

$\square$

4.5 NORMAL MODEL



Another property of $\hat{\beta}$ and $\hat{\sigma}^2$ under normality is that they are sufficient statistics. Intuitively, a statistic is sufficient for a parameter if the statistic summarizes all the information in the sample about the parameter. Sufficiency of $\hat{\beta}$ and $\hat{\sigma}^2$ can be established by the Neyman factorization theorem [see Hogg and Craig (1995, p. 318) or Graybill (1976, pp. 69–70)], which states that $\hat{\beta}$ and $\hat{\sigma}^2$ are jointly sufficient for β and σ^2 if the density $f(y; \beta, \sigma^2)$ can be factored as $f(y; \beta, \sigma^2) = g(\hat{\beta}, \hat{\sigma}^2, \beta, \sigma^2)h(y)$, where $h(y)$ does not depend on β or σ^2 . The following theorem shows that $\hat{\beta}$ and $\hat{\sigma}^2$ satisfy this criterion.

Theorem 7.6c. If y is $N_n(X\beta, \sigma^2 I)$, then $\hat{\beta}$ and $\hat{\sigma}^2$ are jointly sufficient for β and σ^2 .

4.5 NORMAL MODEL



Theorem 7.6d. If y is $N_n(X\beta, \sigma^2 I)$, then $\hat{\beta}$ and s^2 have minimum variance among all unbiased estimators.

Corollary 1. If y is $N_n(X\beta, \sigma^2 I)$, then the minimum variance unbiased estimator of $a'\beta$ is $a'\hat{\beta}$, where $\hat{\beta}$ is the maximum likelihood estimator given in (7.48). $\square$

4.6 R² IN FIXED-x REGRESSION

In (7.39), we have $SSE = \sum_{i=1}^n (y_i - \bar{y})^2 - \hat{\beta}'_1 X'_c y$. Thus the corrected total sum of squares $SST = \sum_i (y_i - \bar{y})^2$ can be partitioned as

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \hat{\beta}'_1 X'_c y + SSE, \quad (7.53)$$

$$SST = SSR + SSE,$$

where $SSR = \hat{\beta}'_1 X'_c y$ is the *regression sum of squares*. From (7.37), we obtain $X'_c y = X'_c X_c \hat{\beta}_1$, and multiplying this by $\hat{\beta}'_1$ gives $\hat{\beta}'_1 X'_c y = \hat{\beta}'_1 X'_c X_c \hat{\beta}_1$. Then $SSR = \hat{\beta}'_1 X'_c y$ can be written as

$$SSR = \hat{\beta}'_1 X'_c X_c \hat{\beta}_1 = (X_c \hat{\beta}_1)' (X_c \hat{\beta}_1). \quad (7.54)$$

In this form, it is clear that SSR is due to $\beta_1 = (\beta_1, \beta_2, \dots, \beta_k)'$.

4.6 R² IN FIXED-x REGRESSION

The proportion of the total sum of squares due to regression is

$$R^2 = \frac{\hat{\beta}'_1 X'_c X_c \hat{\beta}_1}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{SSR}{SST}, \quad (7.55)$$

which is known as the *coefficient of determination* or the *squared multiple correlation*. The ratio in (7.55) is a measure of model fit and provides an indication of how well the x 's predict y .

The partitioning in (7.53) can be rewritten as the identity

$$\sum_{i=1}^n (y_i - \bar{y})^2 = y'y - n\bar{y}^2 = (\hat{\beta}'_1 X'_c y - n\bar{y}^2) + (y'y - \hat{\beta}'_1 X'_c y) = SSR + SSE,$$

which leads to an alternative expression for R^2 :

$$R^2 = \frac{\hat{\beta}'_1 X'_c y - n\bar{y}^2}{y'y - n\bar{y}^2}. \quad (7.56)$$

The positive square root R obtained from (7.55) or (7.56) is called the *multiple correlation coefficient*. If the x variables were random, R would estimate a population multiple correlation (see Section (10.4)).

4.6 R^2 IN FIXED- x REGRESSION

We list some properties of R^2 and R :



1. The range of R^2 is $0 \leq R^2 \leq 1$. If all the $\hat{\beta}_j$'s were zero, except for $\hat{\beta}_0$, R^2 would be 0. (This event has probability 0 for continuous data.) If all the y values fell on the fitted surface, that is, if $y_i = \hat{y}_i$, $i = 1, 2, \dots, n$, then R^2 would be 1.
2. $R = r_{\hat{y}y}$; that is, the multiple correlation is equal to the simple correlation [see (6.18)] between the observed y_i 's and the fitted $\hat{y}_i$'s.
3. Adding a variable x to the model increases (cannot decrease) the value of R^2 .
4. If $\beta_1 = \beta_2 = \dots = \beta_k = 0$, then

$$E(R^2) = \frac{k}{n-1}. \quad (7.57)$$

Note that the $\hat{\beta}_j$'s will not be 0 when the β_j 's are 0.

5. R^2 cannot be partitioned into k components, each of which is uniquely attributable to an x_j , unless the x 's are mutually orthogonal, that is, $\sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{im} - \bar{x}_m) = 0$ for $j \neq m$.
6. R^2 is invariant to full-rank linear transformations on the x 's and to a scale change on y (but not invariant to a joint linear transformation including y and the x 's).

4.7 GENERALIZED LEAST SQUARES



We now consider models in which the y variables are correlated or have differing variances, so that $\text{cov}(y) \neq \sigma^2 \mathbf{I}$. In simple linear regression, larger values of x_i may lead to larger values of $\text{var}(y_i)$. In either simple or multiple regression, if $y_1, y_2, \dots, y_n$ occur at sequential points in time, they are typically correlated. For cases such as these, in which the assumption $\text{cov}(y) = \sigma^2 \mathbf{I}$ is no longer appropriate, we use the model

$$y = \mathbf{X}\beta + \varepsilon, \quad E(y) = \mathbf{X}\beta, \quad \text{cov}(y) = \Sigma = \sigma^2 \mathbf{V}, \quad (7.62)$$

where $\mathbf{X}$ is full-rank and $\mathbf{V}$ is a known positive definite matrix. The usage $\Sigma = \sigma^2 \mathbf{V}$ permits estimation of σ^2 in some convenient contexts (see Examples 7.8.1 and 7.8.2).

The $n \times n$ matrix $\mathbf{V}$ has n diagonal elements and $\binom{n}{2}$ elements above (or below) the diagonal. If $\mathbf{V}$ were unknown, these $\binom{n}{2} + n$ distinct elements could not be estimated from a sample of n observations. In certain applications, a simpler structure for $\mathbf{V}$ is assumed that permits estimation. Such structures are illustrated in Examples 7.8.1 and 7.8.2.

4.7 GENERALIZED LEAST SQUARES Estimation of β and σ^2 when $\text{cov}(y) = \sigma^2 V$



Theorem 7.8a. Let $y = X\beta + \varepsilon$, let $E(y) = X\beta$, and let $\text{cov}(y) = \text{cov}(\varepsilon) = \sigma^2 V$, where X is a full-rank matrix and V is a known positive definite matrix. For this model, we obtain the following results:

- (i) The best linear unbiased estimator (BLUE) of β is

$$\hat{\beta} = (X'V^{-1}X)^{-1}X'V^{-1}y. \quad (7.63)$$

- (ii) The covariance matrix for $\hat{\beta}$ is

$$\text{cov}(\hat{\beta}) = \sigma^2 (X'V^{-1}X)^{-1}. \quad (7.64)$$

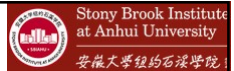
- (iii) An unbiased estimator of σ^2 is

$$s^2 = \frac{(y - X\hat{\beta})'V^{-1}(y - X\hat{\beta})}{n - k - 1} \quad (7.65)$$

$$= \frac{y'[V^{-1} - V^{-1}X(X'V^{-1}X)^{-1}X'V^{-1}]y}{n - k - 1}, \quad (7.66)$$

where $\hat{\beta}$ is as given by (7.63).

4.7 GENERALIZED LEAST SQUARES



PROOF. We prove part (i). For parts (ii) and (iii), see Problems (7.32) and (7.33).

1. Since V is positive definite, there exists an $n \times n$ nonsingular matrix P such that $V = PP'$ (see Theorem 2.6c). Multiplying $y = X\beta + \varepsilon$ by P^{-1} , we obtain $P^{-1}y = P^{-1}X\beta + P^{-1}\varepsilon$, for which $E(P^{-1}\varepsilon) = P^{-1}E(\varepsilon) = P^{-1}0 = 0$ and

$$\begin{aligned} \text{cov}(P^{-1}\varepsilon) &= P^{-1}\text{cov}(\varepsilon)(P^{-1})' \quad [\text{by (3.44)}] \\ &= P^{-1}\sigma^2 V(P^{-1})' = \sigma^2 P^{-1}PP'(P^{-1})^{-1} = \sigma^2 I. \end{aligned}$$

Thus the assumptions for Theorem 7.3d are satisfied for the model $P^{-1}y = P^{-1}X\beta + P^{-1}\varepsilon$, and the least-squares estimator $\hat{\beta} = [(P^{-1}X)'(P^{-1}X)]^{-1}(P^{-1}X)'P^{-1}y$ is BLUE. Using Theorems 2.2b and 2.5b, this can be written as

$$\begin{aligned} \hat{\beta} &= [X'(P^{-1})'P^{-1}X]^{-1}X'(P^{-1})'P^{-1}y \\ &= [X'(P')^{-1}P^{-1}X]^{-1}X'(P')^{-1}P^{-1}y \quad [\text{by (2.48)}] \\ &= [X'(PP')^{-1}X]^{-1}X'(PP')^{-1}y \quad [\text{by (2.49)}] \\ &= (X'V^{-1}X)^{-1}X'V^{-1}y. \end{aligned}$$

4.7 GENERALIZED LEAST SQUARES

Note that since $\mathbf{X}$ is full-rank, $\mathbf{X}'\mathbf{V}^{-1}\mathbf{X}$ is positive definite (see Theorem 2.6b). The estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{y}$ is usually called the *generalized least-squares* estimator. The same estimator is obtained under a normality assumption.

Theorem 7.8b. If $\mathbf{y}$ is $N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{V})$, where $\mathbf{X}$ is full-rank and $\mathbf{V}$ is a known positive definite matrix, where $\mathbf{X}$ is $n \times (k + 1)$ of rank $k + 1$, then the maximum likelihood estimators for $\boldsymbol{\beta}$ and σ^2 are

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{y},$$

$$\hat{\sigma}^2 = \frac{1}{n}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})'\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}).$$

$$\text{Ehat}\{\boldsymbol{\beta}\}=? \text{Cov}(\text{hat}\{\boldsymbol{\beta}\})=? \text{Ehat}\{\sigma^2\}=?$$

4.7 GENERALIZED LEAST SQUARES

PROOF. The likelihood function is

$$L(\boldsymbol{\beta}, \sigma^2) = \frac{1}{(2\pi)^{n/2}|\sigma^2\mathbf{V}|^{1/2}} e^{-(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})'(\sigma^2\mathbf{V})^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})/2}.$$

By (2.69), $|\sigma^2\mathbf{V}| = (\sigma^2)^n|\mathbf{V}|$. Hence

$$L(\boldsymbol{\beta}, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}|\mathbf{V}|^{1/2}} e^{-(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})'\mathbf{V}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})/2\sigma^2}.$$

The results can be obtained by differentiation of $\ln L(\boldsymbol{\beta}, \sigma^2)$ with respect to $\boldsymbol{\beta}$ and with respect to σ^2 . □

4.7 GENERALIZED LEAST SQUARES



Example 7.3.1a

```
y<-c(2, 3, 2, 7, 6, 8, 10, 7, 8, 12, 11,14)
x<- matrix(c(rep(1,12),0,2,2,2,4,4,4,6,6,6,8,8,2,6,7,5,9,8,7,10,11,9,15,3),12,3)
x1<-x[,2]
x2<-x[,3]
lm2<-lm(y~x1+x2)
summary(lm2)
```

Call:
lm(formula = y ~ x1 + x2)

Residuals:

	Min	1Q	Median	3Q	Max
	-2.3223	-1.0808	-0.5592	1.4268	2.7936

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.3288	1.4160	2.351	0.043242 *
x1	1.5893	0.2822	5.631	0.000321 ***
x2	-0.3542	0.2015	-1.758	0.112589

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.934 on 9 degrees of freedom
Multiple R-squared: 0.796, Adjusted R-squared: 0.7506
F-statistic: 17.56 on 2 and 9 DF, p-value: 0.0007827

4.7 GENERALIZED LEAST SQUARES



- 7.53 When gasoline is pumped into the tank of a car, vapors are vented into the atmosphere. An experiment was conducted to determine whether y , the amount of vapor, can be predicted using the following four variables based on initial conditions of the tank and the dispensed gasoline:

x_1 = tank temperature ($^{\circ}\text{F}$)
 x_2 = gasoline temperature ($^{\circ}\text{F}$)
 x_3 = vapor pressure in tank (psi)
 x_4 = vapor pressure of gasoline (psi)

The data are given in Table 7.3 (Weisberg 1985, p. 138).

TABLE 7.3 Gas Vapor Data

y	x_1	x_2	x_3	x_4	y	x_1	x_2	x_3	x_4
29	33	53	3.32	3.42	40	90	64	7.32	6.70
24	31	36	3.10	3.26	46	90	60	7.32	7.20
26	33	51	3.18	3.18	55	92	92	7.45	7.45
22	37	51	3.39	3.08	52	91	92	7.27	7.26
27	36	54	3.20	3.41	29	61	62	3.91	4.08
21	35	35	3.03	3.03	22	59	42	3.75	3.45
33	59	56	4.78	4.57	31	88	65	6.48	5.80
34	60	60	4.72	4.72	45	91	89	6.70	6.60
32	59	60	4.60	4.41	37	63	62	4.30	4.30
34	60	60	4.53	4.53	37	60	61	4.02	4.10
20	34	35	2.90	2.95	33	60	62	4.02	3.89
36	60	59	4.40	4.36	27	59	62	3.98	4.02
34	60	62	4.31	4.42	34	59	62	4.39	4.53
23	60	36	4.27	3.94	19	37	35	2.75	2.64
24	62	38	4.41	3.49	16	35	35	2.59	2.59
32	62	61	4.39	4.39	22	37	37	2.73	2.59

4.7 GENERALIZED LEAST SQUARES

```

datagas<- read.table("D:/2024线性模型/GasVaporData.txt",header=T) # Load data
dim(datagas)
y1<-datagas$y
y2<-datagas$y.1
y<-c(y1,y2)
z1<-datagas$x1
z11<-datagas$x1.1
x1<-c(z1,z11)
z2<-datagas$x2
z22<-datagas$x2.1
x2<-c(z2,z22)
z3<-datagas$x3
z33<-datagas$x3.1
x3<-c(z3,z33)
z4<-datagas$x4
z44<-datagas$x4.1
x4<-c(z4,z44)
lm3<-lm(y~x1+x2+x3+x4)
summary(lm3)

```

```

> lm3<-lm(y~x1+x2+x3+x4)
> summary(lm3)

Call:
lm(formula = y ~ x1 + x2 + x3 + x4)

Residuals:
    Min       1Q   Median       3Q      Max
-5.586 -1.221 -0.118  1.320  5.106

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.01502    1.86131   0.545  0.59001
x1          -0.02861    0.09060  -0.316  0.75461
x2           0.21582    0.06772   3.187  0.00362 **
x3          -4.32005    2.85097  -1.515  0.14132
x4           8.97489    2.77263   3.237  0.00319 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.73 on 27 degrees of freedom
Multiple R-squared:  0.9261,    Adjusted R-squared:  0.9151

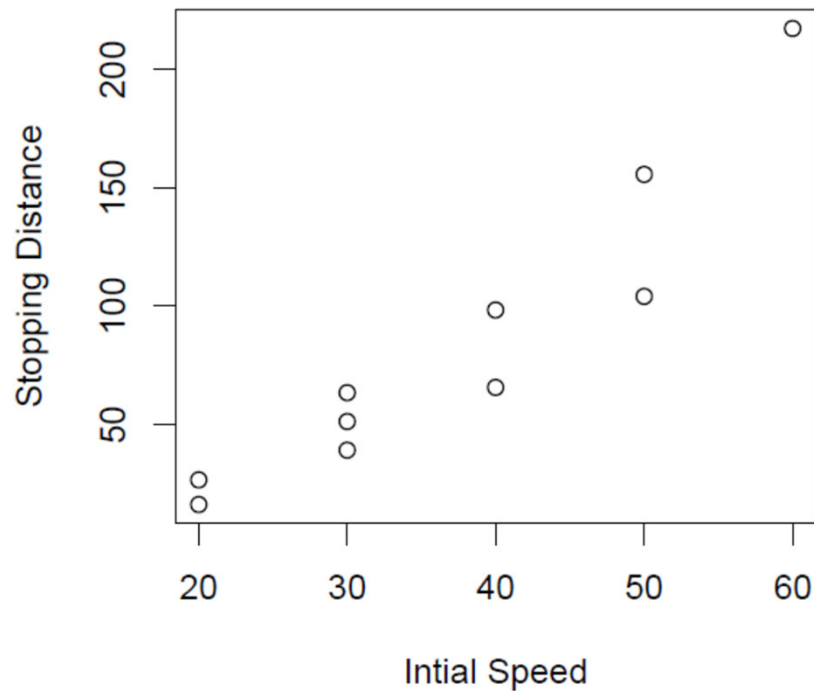
```

12.1 Nonlinear Relations and Transformations

Example 12.1.1 Relationship between the initial speed (miles per hour) and stopping distance (feet). 10 cars are tested.

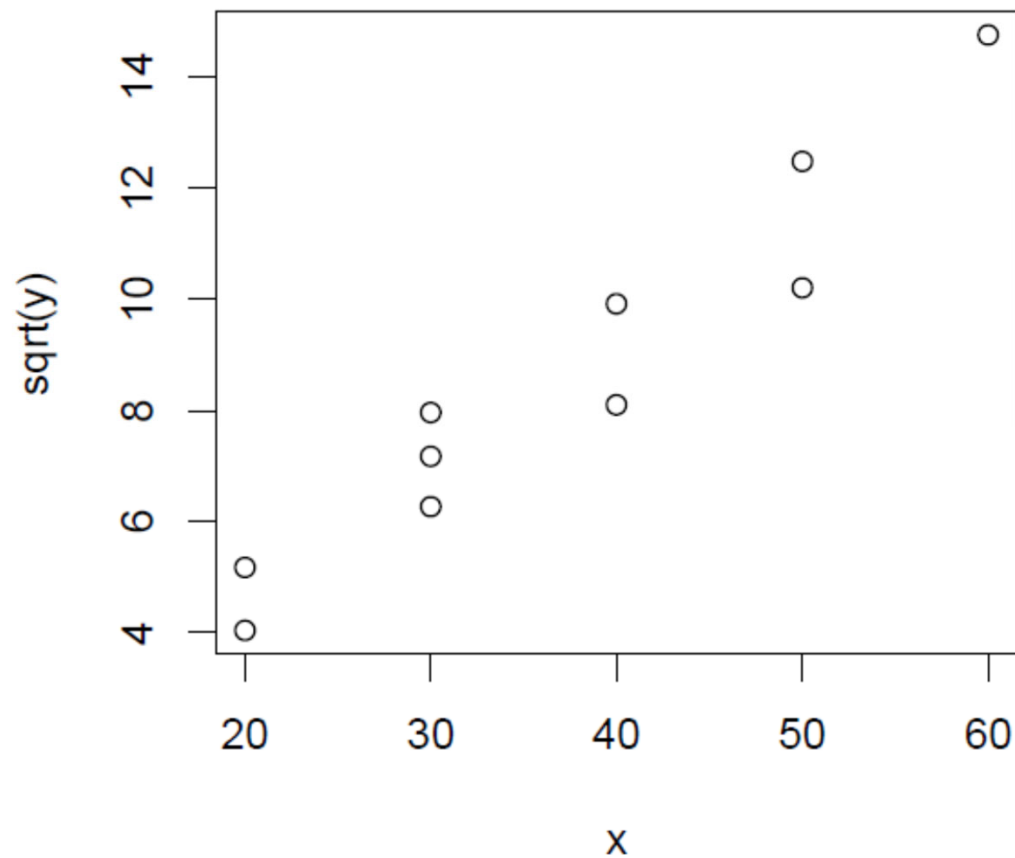
Initial speed (x)	20	20	30	30	30	40	40	50	50	60
Stopping distance(y)	16.3	26.7	39.2	63.5	51.3	98.4	65.7	104.1	155.6	217.2

Nonlinear



Example 12.1 (continued) Transformed data

x	20	20	30	30	30	40	40	50	50	60
$y' = \sqrt{y}$	4.037	5.167	6.261	7.969	7.162	9.920	8.106	10.203	12.474	14.738



x	20	20	30	30	30	40	40	50	50	60
$y' = \sqrt{y}$	4.037	5.167	6.261	7.969	7.162	9.920	8.106	10.203	12.474	14.738

$$y'_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$\bar{x} = 37, \bar{y}' = 8.604,$$

$$S_{xx} = \sum_{i=1}^n x_i^2 - n(\bar{x})^2 = 1610, \quad S_{y'y'} = \sum_{i=1}^n y'^2_i - n(\bar{y}')^2 = 97.773$$

$$S_{xy'} = \sum_{i=1}^n x_i y'_i - n \bar{x} \bar{y}' = 381.621$$

$$\hat{\beta}_1 = \frac{S_{xy'}}{S_{xx}} = \frac{381.621}{1610} = 0.237,$$

$$\hat{\beta}_0 = \bar{y}' - \hat{\beta}_1 \bar{x} = 8.604 - (0.237)(37) = -0.167$$

$$\hat{y}' = -0.167 + 0.237x$$

$$R^2 = \frac{S_{xy'}^2}{S_{xx} S_{y'y'}} = \frac{(381.621)^2}{1610 \cdot 97.773} = 0.925$$



Output in R

```
>x=c(20,20,30,30,30,40,40,50,50,60)
>y=c(16.3,26.7,39.2,63.5,51.3,98.4,65.7,104.1,155.6,217.2)
>squ=sqrt(y)
>fit=lm(squ ~ x)
>summary(fit)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.4821	-0.6468	0.4055	0.6630	1.0243

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.16653	0.93225	-0.179	0.863
x	0.23703	0.02383	9.945	8.84e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9563 on 8 degrees of freedom

Multiple R-squared: 0.9252, Adjusted R-squared: 0.9158

F-statistic: 98.91 on 1 and 8 DF, p-value: 8.843e-06

Some nonlinear models and their linearizing transformations

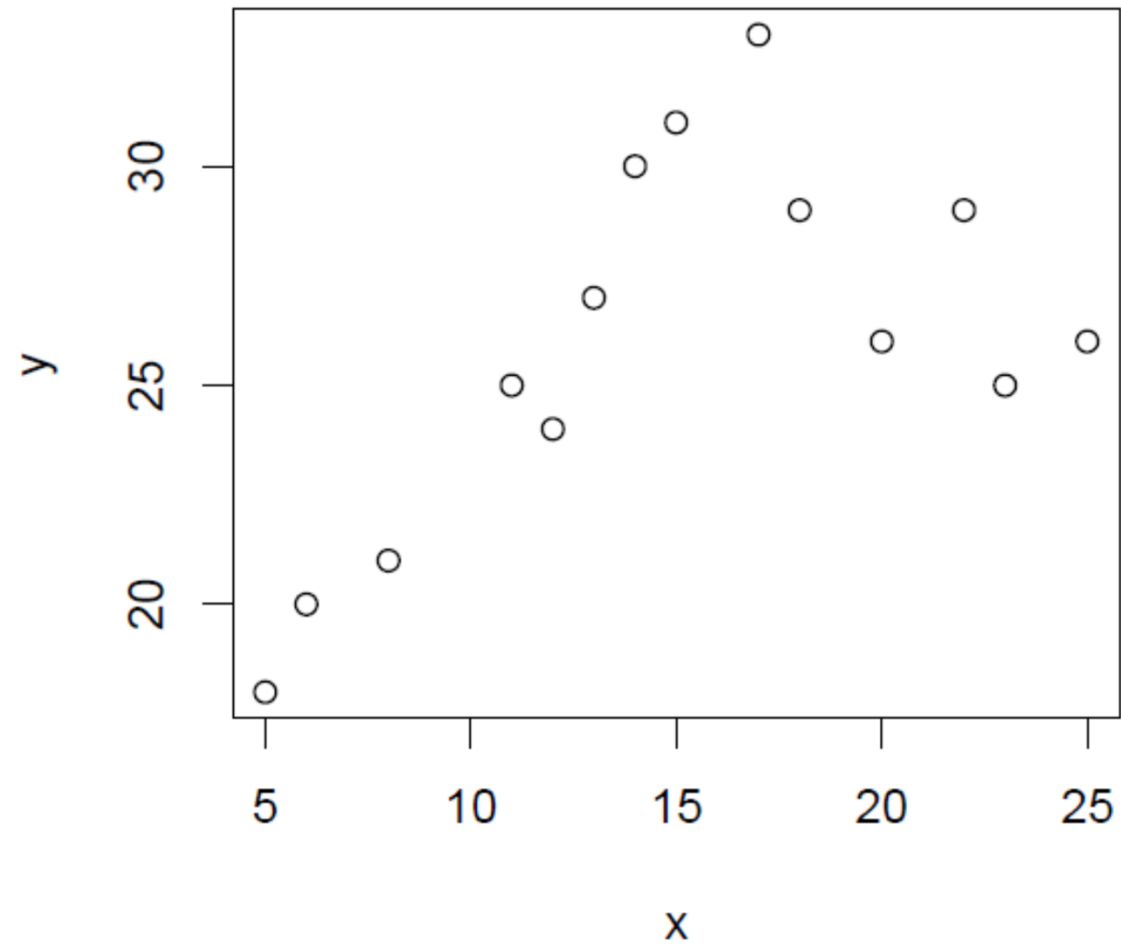
			Linear form	
Function	Transformation		$y' = \beta_0 + \beta_1 x'$	
$y = ae^{bx}$	$y' = \ln y$	$x' = x$	$\beta_0 = \ln a$	$\beta_1 = b$
$y = ax^b$	$y' = \log y$	$x' = \log x$	$\beta_0 = \log a$	$\beta_1 = b$
$y = \frac{1}{a + bx}$	$y' = \frac{1}{y}$	$x' = x$	$\beta_0 = a$	$\beta_1 = b$
$y = a + b\sqrt{x}$	$y' = y$	$x' = \sqrt{x}$	$\beta_0 = a$	$\beta_1 = b$

12.2 Polynomial Regression

Example 12.2.1 Yield of 14 equal-sized plots of tomato plantings for different amounts of fertilizer

Plot	Yields, y (in bushels)	Amount of fertilizer, x (in pounds per plot)
1	24	12
2	18	5
3	31	15
4	33	17
5	26	20
6	30	14
7	20	6
8	25	23
9	25	11
10	27	13
11	21	8
12	29	18
13	29	22
14	26	25

Example 12.2.1 (continued)



Output in R

```
> x=c(12,5,15,17,20,14,6,23,11,13,8,18,22,25)
> y=c(24,18,31,33,26,30,20,25,25,27,21,29,29,26)
> xx=x^2
> fit=lm(y~x+xx) # not fit=lm(y~x+x^2)
> summary(fit)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.9646	-1.2584	0.2213	1.1291	3.6970

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.70311	3.30049	1.728	0.111926
x	2.69234	0.48142	5.592	0.000162 ***
xx	-0.07671	0.01586	-4.835	0.000523 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.12 on 11 degrees of freedom

Multiple R-squared: 0.794, Adjusted R-squared: 0.7566

F-statistic: 21.2 on 2 and 11 DF, p-value: 0.0001682

Polynomial Regression

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i$$

In general,

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_k x_i^k + \varepsilon_i$$

12.3 Multiple Regression

Example 12.3.1 We are interested in studying on systolic blood pressure y in relation to weight x_1 and age x_2 in a class of males of approximately the same height. From 13 subjects preselected according to weight and age, data are obtained.

Weight (x_1) (in pounds)	Age (x_2)	Blood pressure (y)
152	50	120
183	20	141
171	20	124
165	30	126
158	30	117
161	50	129
149	60	123
158	50	125
170	40	132
153	55	123
164	40	132
190	40	155
185	20	147

Output in R

```
>x1=c(152,183,171,165,158,161,149,158,170,153,164,190,185)
>x2=c(50,20,20,30,30,50,60,50,40,55,40,40,20)
>y=c(120,141,124,126,117,129,123,125,132,123,132,155,147)
>fit=lm(y ~ x1+x2) # or lm(y ~ x1+x2, data=filename)
>summary(fit)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.5929	-1.3530	-0.0946	0.6155	4.3276

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-65.09968	14.94458	-4.356	0.001430 **
x1	1.07710	0.07707	13.975	6.89e-08 ***
x2	0.42541	0.07315	5.815	0.000169 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.509 on 10 degrees of freedom

Multiple R-squared: 0.9577, Adjusted R-squared: 0.9492

F-statistic: 113.1 on 2 and 10 DF, p-value: 1.359e-07

Output in R (continued)

```
>fit=lm(y ~ x1+x2) # or lm(y ~ x1+x2, data=filename)  
>coef(fit)      # or fit$coefficients, model coefficients
```



(Intercept)	x1	x2
-65.0996781	1.0771015	0.4254134

$$\hat{y} = -65.1 + 1.08x_1 + 0.425x_2$$

The mean blood pressure increases by 1.08 if weight (x_1) increases by one pound and age (x_2) remains fixed. Similarly, a 1-year increase in age with the weight held fixed will increase the mean blood pressure by 0.425.

```
>vcov(fit)      # covariance matrix for the parameters  
>influence(fit) # regression diagnostics
```

$$\hat{y}_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$$

Hypothesis Testing

Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-65.09968	14.94458	-4.356 0.001430 **
x1	1.07710	0.07707	13.975 6.89e-08 ***
x2	0.42541	0.07315	5.815 0.000169 ***

To test H_0 that a particular coefficient β is 0, the test statistic is

$$T = \frac{\text{Estimated coefficient} - 0}{\text{Estimated se}}, \text{ df} = n - 3 = 10.$$

For $H_0: \beta_1 = 0$ vs. $H_1: \beta_1 > 0$ with $\alpha = 0.01$, the rejection region is $t > t_{\alpha,10} = t_{0.01,10} = 2.764$

$$t = \frac{\hat{\beta}_1}{S_{\hat{\beta}_1}} = \frac{1.0771}{0.07707} = 13.98 > 2.764$$

Reject H_0 .

For $H_0: \beta_2 = 0$ vs. $H_1: \beta_2 > 0$ with $\alpha = 0.01$

$$t = \frac{\hat{\beta}_2}{S_{\hat{\beta}_2}} = \frac{0.42541}{0.07315} = 5.82 > 2.764$$

Reject H_0 .

Output in R (continued)

A $100(1 - \alpha)\%$ CI for a coefficient β is

Estimated coefficient $\pm t_{\frac{\alpha}{2}, 10} \times (\text{estimated se})$.

e.g., a 99% CI for β_1 is $1.077 \pm 3.169(0.077) = 1.077 \pm 0.244$
or (0.833, 1.321).

```
>confint(fit, level=0.99) # CIs for parameters. Default: 0.95
```

	0.5 %	99.5 %
(Intercept)	-112.4631128	-17.736243
x1	0.8328386	1.321364
x2	0.1935738	0.657253





Output in R (continued)

>fitted(fit) # predicted values
>anova(fit) # ANOVA table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
x1	1	1211.01	1211.01	192.433	7.393e-08 ***
x2	1	212.83	212.83	33.819	0.0001694 ***
Residuals	10	62.93	6.29		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Source	DF	SS	MS	F	p-value
Regression	2	1423.84	$\frac{1423.84}{2} = 711.92$	$\frac{711.92}{6.29} = 113.13$	0.0000
Error	10	62.93	$\frac{62.93}{10} = 6.29$		
Total	12	1486.77			

$\downarrow \frac{SSR}{2} \sim F_{2, n-3}$
 $\frac{SSE}{(n-3)}$

$\frac{SSR}{\sigma^2} \sim \chi^2_2$, $\frac{SSE}{\sigma^2} \sim \chi^2_{n-3}$

Output (continued)

Source	DF	SS	MS	F	p-value
Regression	2	1423.84	$\frac{1423.84}{2} = 711.92$	$\frac{711.92}{6.29} = 113.13$	0.0000
Error	10	62.93	$\frac{62.93}{10} = 6.29$		
Total	12	1486.77			

$$R^2 = \frac{SSR}{SST} = \frac{1423.84}{1486.77} = 0.958$$

95.8% of the variability of y is explained by the fitted multiple regression of y on x_1 and x_2 .

$$MSE = s^2 = 6.293, \hat{\sigma} = s = \sqrt{6.293} = 2.509$$

>sigma(fit) # Residual standard error σ
2.50861

The error sd is estimated by $\sigma = 2.509$ with
 $df = n - (\text{\#variables}) - 1 = 13 - 2 - 1 = 10$

Output (continued)

Source	DF	SS	MS	F	p-value
Regression	2	1423.84	$\frac{1423.84}{2} = 711.92$	$\frac{711.92}{6.29} = 113.13$	0.0000
Error	10	62.93	$\frac{62.93}{10} = 6.29$		
Total	12	1486.77			

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2 = SSR + SSE$$

$$SSR = SST - SSE$$

$$\sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \bar{y})^2 - \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$H_0: \beta_1 = \beta_2 = \cdots = \beta_k = 0 \text{ vs. } H_1: \beta_i \neq 0 \text{ for at least one } i$$

$$\text{Test statistic: } F = \frac{SSR/k}{SSE/[n-(k+1)]} = \frac{MSR}{MSE}$$

$$\text{Rejection region: } F > F_{\alpha, k, n-k-1} = F_{0.05, 2, 10} = 4.10$$

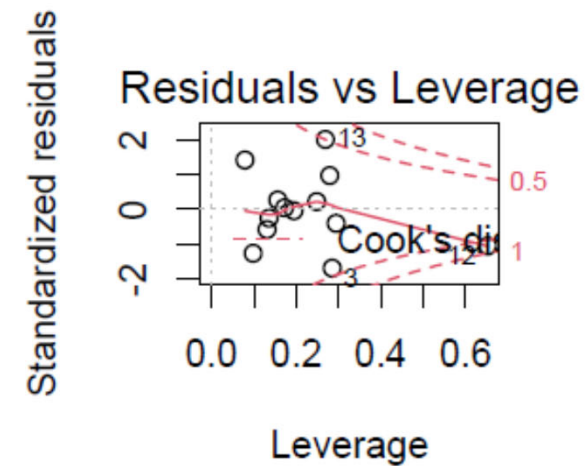
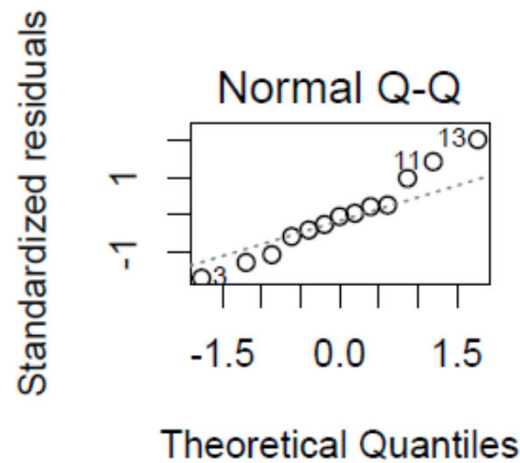
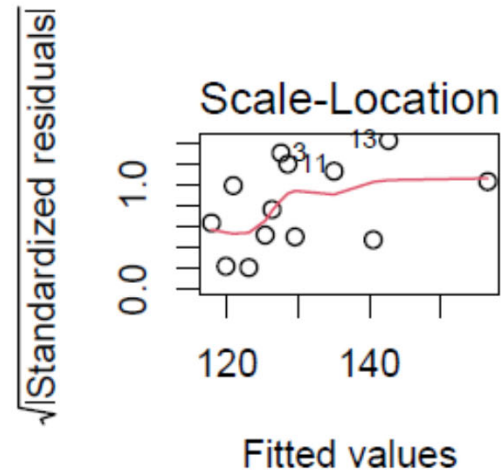
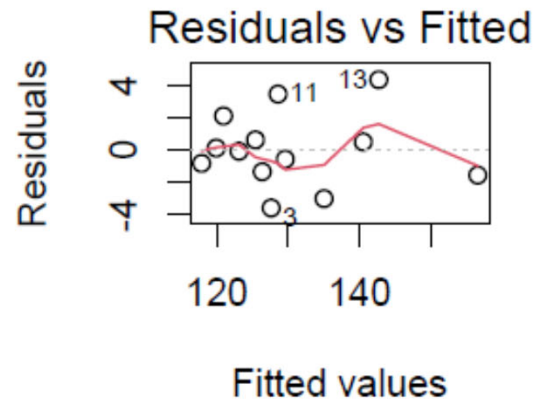
$$F = \frac{711.92}{6.29} = 113.13 > 4.10 \Rightarrow \text{Reject } H_0$$

1.2.5



Diagnostic plots

`layout(matrix(c(1,2,3,4),2,2)) #optional 4 graphs/page`
`plot(fit)`



Output in R (continued)

```
>fit1=lm(y ~ x1)
```



```
>anova(fit, fit1)          # ANOVA table
```

Model 1: $y \sim x_1 + x_2$

Model 2: $y \sim x_1$

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	10	62.931				
2	11	275.761	-1	-212.83	33.819	0.0001694 ***

Adding interaction terms: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \varepsilon$

```
>fit2=lm(y ~ (x1+x2)^2)
```

```
>summary(fit2)
```

Handwritten red text: $\text{fit2} = \text{lm}(y \sim x_1 + x_2 + x_1 * x_2)$

Estimate Std. Error t value Pr(>|t|)

(Intercept) -1.140e+02 3.123e+01 -3.650 0.00531 **

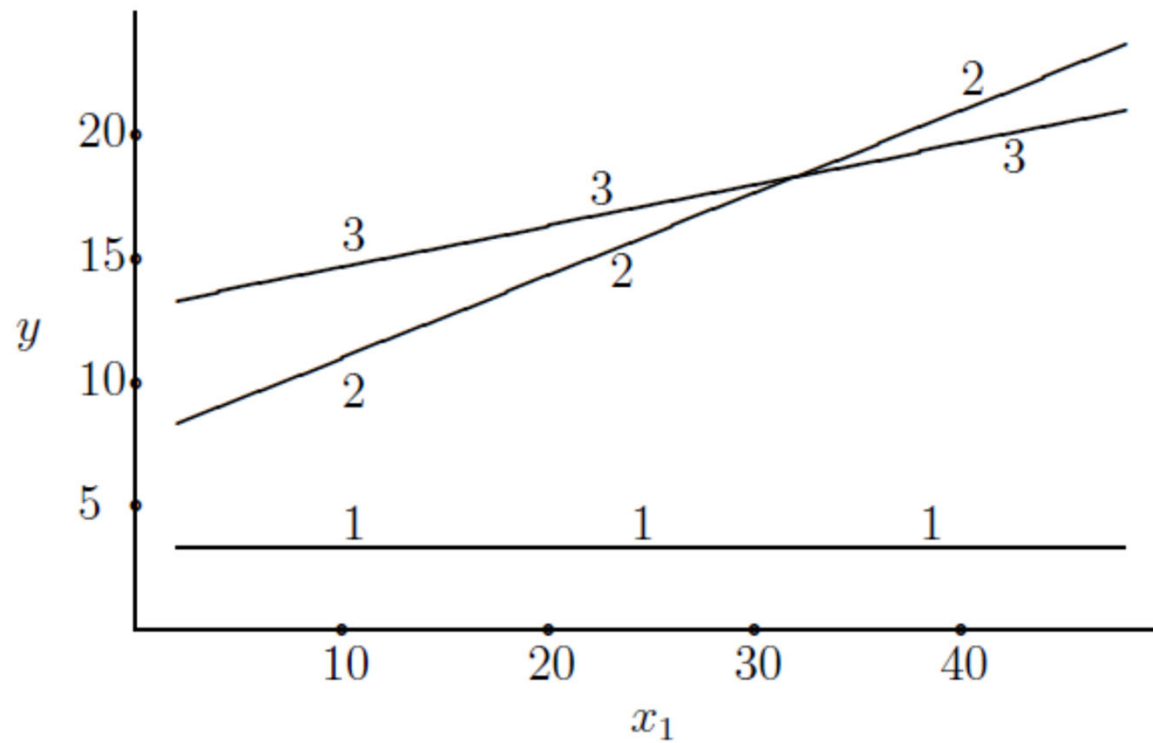
x1 1.368e+00 1.811e-01 7.552 3.5e-05 ***

x2 1.685e+00 7.269e-01 2.318 0.04561 *

x1:x2 -7.621e-03 4.379e-03 -1.740 0.11578

Interactions

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \varepsilon$$



Scatterplot of y vs. x_1 at three levels of x_2

12.4 Least Squares Estimation

$$X = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix}_{n \times (k+1)} \quad Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}_{n \times 1} \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}_{k+1}$$

$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}_{n \times 1}$

$$L(D, \beta) = \|Y - X\beta\|^2 = (Y - X\beta)^T (Y - X\beta) \\ = Y^T Y - Y^T X\beta - \beta^T X^T Y + \beta^T X^T X\beta$$

$$Y = X\beta + \varepsilon$$

$$\frac{\partial L(D, \beta)}{\partial \beta} = -2X^T Y + 2X^T X\beta = 0$$

$$\Rightarrow \hat{\beta} = \underline{(X^T X)^{-1} X^T Y}$$

$$\text{Cov}(Y) = \sigma^2 I_n$$

I_n is $n \times n$ identity matrix

要知 $X^T X$ 可逆

$$\text{Cov}(\hat{\beta}) = (X^T X)^{-1} \sigma^2$$

or $\text{rank}(X^T X) = k+1$

$$\begin{aligned} \text{Cov}(\hat{\beta}) &= (X^T X)^{-1} X^T \text{Cov}(Y) X [(X^T X)^{-1}]^T \\ &= (X^T X)^{-1} X^T \sigma^2 I X (X^T X)^{-1} \\ &= \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1} = \sigma^2 (X^T X)^{-1} \end{aligned}$$

Example 12.3.1 (continued)

$$X = \begin{pmatrix} 1 & 152 & 50 \\ 1 & 183 & 20 \\ 1 & 171 & 20 \\ 1 & 165 & 30 \\ 1 & 158 & 30 \\ 1 & 161 & 50 \\ 1 & 149 & 60 \\ 1 & 158 & 50 \\ 1 & 170 & 40 \\ 1 & 153 & 55 \\ 1 & 164 & 40 \\ 1 & 190 & 40 \\ 1 & 185 & 20 \end{pmatrix}, \quad Y = \begin{pmatrix} 120 \\ 141 \\ 124 \\ 126 \\ 117 \\ 129 \\ 123 \\ 125 \\ 132 \\ 123 \\ 132 \\ 155 \\ 147 \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}$$

$$X^T X = \begin{pmatrix} 13 & 2159 & 505 \\ 2159 & 360639 & 82335 \\ 505 & 82335 & 21925 \end{pmatrix}, \quad (X^T X)^{-1} = \begin{pmatrix} 35.4896 & -0.1811 & -0.1372 \\ -0.1811 & 0.00094 & 0.00063 \\ -0.1372 & 0.00063 & 0.00085 \end{pmatrix}$$

$$X^T Y = \begin{pmatrix} 1694 \\ 282921 \\ 65135 \end{pmatrix}, \quad \hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T Y = \begin{pmatrix} -65.0997 \\ 1.0771 \\ 0.4254 \end{pmatrix} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix}$$

Example 12.3.1 (continued)

In R,

```
y=c(120,141,124,126,117,129,123,125,132,123,132,155,147)
```

```
x0=rep(1,13)
```

```
x1=c(152,183,171,165,158,161,149,158,170,153,164,190,185)
```

```
x2=c(50,20,20,30,30,50,60,50,40,55,40,40,20)
```

```
x=cbind(x0,x1,x2)
```

```
xx=t(x)%*%x
```

```
xy=t(x)%*%y
```

```
beta=solve(xx)%*%xy
```

$$X^T X = \begin{pmatrix} 13 & 2159 & 505 \\ 2159 & 360639 & 82335 \\ 505 & 82335 & 21925 \end{pmatrix}, \quad (X^T X)^{-1} = \begin{pmatrix} 35.4896 & -0.1811 & -0.1372 \\ -0.1811 & 0.00094 & 0.00063 \\ -0.1372 & 0.00063 & 0.00085 \end{pmatrix}$$

$$X^T Y = \begin{pmatrix} 1694 \\ 282921 \\ 65135 \end{pmatrix}, \quad \hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T Y = \begin{pmatrix} -65.0997 \\ 1.0771 \\ 0.4254 \end{pmatrix} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix}$$

Example 12.3.1 (continued)



```
>yhat=x%*%beta  
>SSE=t(y-yhat)%*%(y-yhat)
```

$$\hat{Y} = X\hat{\beta}$$

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = (Y - \hat{Y})^T (Y - \hat{Y}) = 62.93126$$

```
>SSE=as.numeric(SSE)  
>var=(SSE/10)*solve(xx)
```

$$\hat{\sigma}^2 (X^T X)^{-1} = \frac{SSE}{n-3} (X^T X)^{-1} = \begin{pmatrix} 223.340 & -1.1399 & -0.8636 \\ -1.1399 & 0.00594 & 0.00395 \\ -0.8636 & 0.00395 & 0.00535 \end{pmatrix}$$

$$\begin{aligned} \text{se}(\hat{\beta}_0) &= \sqrt{223.34} = 14.94, & \text{se}(\hat{\beta}_1) &= \sqrt{0.00594} = 0.0771, \\ \text{se}(\hat{\beta}_2) &= \sqrt{0.00535} = 0.0731 \end{aligned}$$

```
>sqrt(diag(var))  
x1      x2      x3  
14.94457509 0.07707220 0.07315231
```

$$\frac{\hat{\beta}_2 - \beta_2}{\text{se}(\hat{\beta}_2)} \sim t_{n-3}$$

属性数据

Categorical Predictor Variables

Two types of predictor variables:

- Quantitative variables: numerical values

定量型

 - eg: population size of a city, number of square meters in a house, age
- Categorical variables: variables that take on names or labels

定性数据

 - eg: eye color, gender, marital status

For a categorical variable, create dummy variables.

- For a variable with k categories, create $k - 1$ variables with 0's and 1's.

虚拟变量

k 个属性数据, 要引入 $k-1$ 个虚拟变量

1, 2, 3, 4 属性

1 2 3 4
~~1~~ ~~2~~ ~~3~~ ~~4~~
 d_1, d_2, d_3

分 = 层
 三个类

For example, if $k = 4$,

2	0	0	0
4	0	0	0
1	1	0	0
4	0	0	0
3	0	0	1
2	0	1	0
1	1	0	0
1	1	0	0
4	0	0	0
2	0	1	0
2	0	1	0
3	0	0	1
1	1	0	0
4	0	0	0
3	0	0	1

3个值

变量

d_1, d_2, d_3

0 或 1

Example 12.4.1

Income	Age	Marital Status
\$54000	24	Single
\$59000	27	Single
\$63000	25	Single
\$67000	29	Married
\$75000	37	Married
\$79000	34	Single
\$87000	39	Single
\$92000	58	Divorced
\$108000	54	Married
\$114000	62	Divorced
\$116000	55	Married
\$132000	59	Divorced



Income	Age	Dummy 1	Dummy 2
\$54000	24	1	0
\$59000	27	1	0
\$63000	25	1	0
\$67000	29	0	1
\$75000	37	0	1
\$79000	34	1	0
\$87000	39	1	0
\$92000	58	0	0
\$108000	54	0	1
\$114000	62	0	0
\$116000	55	0	1
\$132000	59	0	0

Variable Selection

- Stepwise regression
- Backward elimination
- Forward selection
- All possible subsets regression
 - AIC (Akaike Information Criterion)
 - BIC (Bayes Information Criterion)



12.5 Logistic Regression

The response is a binary random variable taking on the values 0 & 1.

$$p = P(Y = 1)$$
$$\ln \left(\frac{p(x)}{1 - p(x)} \right) = \beta_0 + \beta_1 x$$
$$p(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}, \quad p(0) = \frac{e^{\beta_0}}{1 + e^{\beta_0}}$$

$$\beta_0 = \beta_1 = 0 \Rightarrow p(x) = \frac{1}{2}$$

$\ln 1 = 0$

$$\text{Odds: } \frac{p(x)}{1 - p(x)} = e^{\beta_0 + \beta_1 x} = e^{\beta_0} (e^{\beta_1})^x$$

An increase of one unit in x results in the odds of the specified event being multiplied by e^{β_1} .

$$\text{Odds ratio: } \frac{p(x_2)/[1-p(x_2)]}{p(x_1)/[1-p(x_1)]} = (e^{\beta_1})^{x_2 - x_1}$$

$$\text{If } x_2 = x_1 + 1, \quad \frac{p(x+1)/[1-p(x+1)]}{p(x)/[1-p(x)]} = e^{\beta_1}$$

Example 12.5.1 (Ott Example 12.22) A study recorded the level of an enzyme, creatinine kinase (CK), for patients who were suspected of having a heart attack. The objective of the study was to assess whether measuring the amount of CK on admission to the hospital was a useful diagnostic indicator of whether patients admitted with a diagnose of a heart attack had really had a heart attack. The enzyme CK was measured in the patients on admission to the hospital. After a period of time, a doctor reviewed the records of these patients to decide which of the 360 patients had actually had a heart attack. The data are given in the table on the next page with the CK values given as the midpoint of the range of values in each of 13 classes of values.

Example 12.5.1 (continued)

CK Value	Number of Patients with Heart Attack	Number of Patients without Heart Attack
20	2	88
60	13	26
100	30	8
140	30	5
180	21	0
220	19	1
260	18	1
300	13	1
340	19	0
380	15	0
420	7	0
460	8	0
500	35	0

Example 12.5.1 (continued)



```
x=c(rep(20,90),rep(60,39),rep(100,38),rep(140,35),rep(180,21),rep(220,20),rep(260,19),  
rep(300,14),rep(340,19),rep(380,15),rep(420,7),rep(460,8),rep(500,35))
```

```
y=c(rep(1,2),rep(0,88),rep(1,13),rep(0,26),rep(1,30),rep(0,8),rep(1,30),rep(0,5),rep(1,21  
,rep(1,19),rep(0,1),rep(1,18),rep(0,1),rep(1,13),rep(0,1),rep(1,19),rep(1,15),rep(1,7),rep  
(1,8),rep(1,35))
```

```
logit=glm(y~x,family=binomial)  
summary(logit)
```

```
logit$coefficients
```

CK Value	Number of Patients with Heart Attack	Number of Patients without Heart Attack
20	2	88
60	13	26
100	30	8
140	30	5
180	21	0
220	19	1
260	18	1
300	13	1
340	19	0
380	15	0
420	7	0
460	8	0
500	35	0

Example 12.5.1 (continued)

Deviance Residuals:

Min	1Q	Median	3Q	Max
-3.8739	-0.4317	0.0061	0.2704	2.1998

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.028360	0.366971	-8.252	<2e-16 ***
x	0.035104	0.004081	8.602	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 470.92 on 359 degrees of freedom
Residual deviance: 187.77 on 358 degrees of freedom
AIC: 191.77

Number of Fisher Scoring iterations: 7

Example 12.5.1 (continued)

a) Is CK level significantly related to the probability of a heart attack?

$H_0: \beta_1 = 0$ vs. $H_1: \beta_1 \neq 0$. Since $p\text{-value} \approx 0$, yes, there is a significant evidence that CK is related to the probability of having had a heart attack.

b) Obtain the estimated coefficients β_0 and β_1 .

$$\beta_0 = -3.0284, \beta_1 = 0.0351 > 0$$

Patients having higher levels of CK are associated with a larger probability that a heart attack had occurred. The odds of having had a heart attack for a patient with a CK level of $x + 1$ is $e^{0.0351} = 1.0357$ times the odds for a patient having a CK level of x .

`>exp(coef(logit))`

c) Construct the estimated probability of having had a heart attack as a function of CK level. What is it for a patient having a CK level of 1.40?

$$\hat{p}(\text{CK}) = \frac{e^{-3.0284+0.0351\text{CK}}}{1 + e^{-3.0284+0.0351\text{CK}}}$$
$$\hat{p}(1.40) = \frac{e^{-3.0284+0.0351(1.40)}}{1 + e^{-3.0284+0.0351(1.40)}} = 0.868$$

$$\hat{p}(x) = \dots$$
$$\hat{p}(1.40)$$

Example 12.5.2 (Ott Example 12.23) A study was conducted to examine the extent to which red blood cells settle out of suspension in blood plasma; erythrocyte sedimentation rate (ESR) is related to two proteins that are present in blood plasma. Individuals are classified as healthy ($ESR < 20$) or unhealthy ($ESR \geq 20$). The two blood plasma proteins are fibrinogen (x_1) and γ -globulin (x_2) and are measured on each of the patients in units of grams/liter. The researchers wanted to determine the strength of the relationship between the probability of determining a patient was unhealthy ($ESR \geq 20$) and the levels of the two plasma proteins. The data are given on the next page with Hth (1 =unhealthy, 0 =healthy), Fib (level of fibrinogen), and Gam (level of γ -globulin).

Example 12.5.2 (continued)

Fib	2.52	2.46	2.29	3.15	2.88	2.29	2.99	2.38
Gam	38	36	36	36	30	31	36	37
Hth	0	0	0	0	0	0	0	1
Fib	2.56	3.22	2.35	3.53	2.65	2.15	3.32	2.23
Gam	31	38	29	46	46	31	35	37
Hth	0	0	0	1	0	0	0	0
Fib	2.19	2.21	5.06	2.68	2.09	2.54	2.18	2.67
Gam	33	37	37	34	44	28	31	39
Hth	0	0	1	0	1	0	0	0
Fib	3.41	3.15	3.34	2.60	2.28	3.93	2.60	3.34
Gam	37	39	32	38	36	32	41	30
Hth	0	0	1	0	0	1	0	0

Example 12.5.2 (continued)

```
Fib=c(2.52,2.46,2.29,3.15,2.88,2.29,2.99,2.38,2.56,3.22,2.35,3.53,2.65,2.15,3.32,2.23,2.19,2.21,5.06,2.68,2.09,2.54,2.18,2.67,3.41,3.15,3.34,2.60,2.28,3.93,2.60,3.34)
```

```
Gam=c(38,36,36,36,30,31,36,37,31,38,29,46,46,31,35,37,33,37,37,34,44,28,31,39,37,39,32,38,36,32,41,30)
```

```
Hth=c(rep(0,7),1,0,0,0,1,rep(0,6),1,0,1,rep(0,5),1,0,0,1,0,0)
```

```
>logit=glm(Hth~Fib+Gam,family=binomial)
```

```
>confint(logit)
```

	2.5 %	97.5 %
(Intercept)	-27.27978361	-3.1496138
Fib	0.33941838	4.2904709
Gam	-0.06617422	0.4266588

Example 12.5.2 (continued)

```
>summary(logit)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-0.9683	-0.6122	-0.3458	-0.2116	2.2636

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-12.7921	5.7963	-2.207	0.0273 *
Fib	1.9104	0.9710	1.967	0.0491 *
Gam	0.1558	0.1195	1.303	0.1925

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 30.885 on 31 degrees of freedom
Residual deviance: 22.971 on 29 degrees of freedom
AIC: 28.971

Number of Fisher Scoring iterations: 5

Example 12.5.2 (continued)

a) Are the levels of the two plasma proteins related to the probability that a patient has an unhealthy level of ESR?

$H_0: \beta_1 = 0$ vs. $H_1: \beta_1 \neq 0$: $p\text{-value} \approx 0.0491$

$H_0: \beta_2 = 0$ vs. $H_1: \beta_2 \neq 0$: $p\text{-value} \approx 0.1925$

The level of fibrinogen is significantly related to the unhealthy level of ESR, but the level of γ -globulin is not at $\alpha = 0.05$.

b) Obtain the estimated coefficients β_0 , β_1 and β_2 .

$$\beta_0 = -12.7921, \beta_1 = 1.9104, \beta_2 = 0.1558$$

c) Construct the estimated probability that a patient has an unhealthy level of ESR as a function of the two predictor variables.

$$\hat{p}(x_1, x_2) = \frac{e^{-12.7920 + 1.9104 x_1 + 0.1558 x_2}}{1 + e^{-12.7920 + 1.9104 x_1 + 0.1558 x_2}}$$